



PROJECT REPORT

Zero-Touch-Pad

Name: Vethra M

Registration Number: 25MIB10011

Branch: Int M.tech AI Bioinformatics

Year: 2026-27

Course Title: Fundamentals in AI & ML

Course Code: CSA2001

Slot: A11+A12+A13

Faculty Name: Professor Atul Onkarrao Thakare

Class No: BL2026270100657,

1. Introduction to the Problem

Today, all the things are about accessibility. I noticed that people working in kitchen or garage or just anyone with dirty hands or unhygienic conditions cannot and should not be using the touch as it will get dirty and bad for others who will be using it.

This means that devices which require physical contact can be:

- Inconvenient when hands are occupied or dirty
- Unhygienic in shared environments
- Difficult for users with certain physical limitations

There is a need for a system that allows users to interact with their computer **without touching any hardware**, while still maintaining usability and accuracy.

2. Objective

The main objectives of this project are:

- To design system that detects hand gestures currently
- To track gestures which does actions
- To create a smooth and easy to use app for people
- To make sure the app works well with only a camera

3. Approach and Method to do this

The solution is based on **computer camera use and gesture recognition**.

Step-by-step working:

- The camera on device sees live video input
- Each frame is processed using MediaPipe installed before
- MediaPipe sees the hand and checks and marks **21 landmark points**
- The app checks which fingers are up or down to do actions
- It calculates distances between fingers for different actions
- Based on predefined conditions, gestures are recognized
- PyAutoGUI performs the corresponding system action

4. Features

1. Cursor Control -> Index finger
2. Left click -> Index + Thumb touch
3. Right Click -> Middle + Thumb touch
4. Show Desktop -> Palm Down
5. Close App -> Make fist for 3 seconds
6. Volume -> Distance between Thumb and Little finger
7. Scroll -> First 2 fingers up or down
8. Drag -> long left click
9. Screenshot -> First 3 fingers up

5. Key Design Decisions

While developing this project, several important decisions were made:

1. Using MediaPipe

I chose mediapipe because it gives accurate and fast hand track option with already made models, making it less complex to make.

2. Gesture Simplicity

Gestures are simple so:

- They are easy to perform
- They reduce confusion
- They improve accuracy

3. One-Hand Only

Only one hand is used to:

- Avoid making it very complex
- Make sure no fake detects
- Make performance better

4. Use of Time Thresholds

Time delays were added to:

- Prevent accidental triggers
- Make gestures more intentional

5. Smooth Cursor Movement

Cursor movement was smoothened using interpolation to:

- Avoid jitter
- Improve user experience

6. Challenges Faced

During development, several challenges were encountered:

1. Gesture Conflicts

Some gestures overlapped (e.g., cursor movement vs scrolling), which caused unintended actions.

Solution:

Separate conditions and gesture priority were introduced.

2. Sensitivity Issues

Small hand movements caused unintended clicks or scrolling.

Solution:

Distance thresholds and conditions were fine-tuned.

3. Lighting Conditions

The system did not work well in poor lighting.

Solution:

Testing was done under proper lighting, and instructions were added for users.

4. Drag Detection

Initially, drag and click were confused.

Solution:

A time-based hold condition was added for drag.

5. Performance and Lag

High sensitivity caused jitter in cursor movement.

Solution:

Smoothening factor was introduced.

7. Results

The final system successfully does the following:

- Tracks hand currently time
- Accurately recognises all hand actions
- Smooth movement for cursor
- Reliable implement of actions
- The app works very good under normal light and gives a correct response.

8. Limitations

All though it works well, it has some shortcomings:

- Does not work with bad lighting
- Accuracy may reduce with complex backgrounds
- Needs a camera for app to work
- Fast movement can reduce detection accuracy

9. Future Improvements

This project can be improved further by:

- Multi-hand support
- Custom gesture mapping
- Better accuracy
- Better speed
- Can be used in phone
- Include voice
- Include ASL

10. What I Learnt

- This project helped me learn:
- Practical application of computer vision
- How gesture recognition works
- Real-time video processing
- Handling user interaction without physical input
- Debugging and improving system performance

- Importance of user experience in design

It also improved my problem-solving skills and understanding of how software can be used to solve real-world problems.

11. Conclusion

The Zero Touch Pad shows how technology should be used to make touch-free methods. It gives a correct and usable solution for real-world conditions where usual input devices are not useful.

This project shows how computer camera can be used to create a touch-less user interface. It gives a practical solution for real-world problem when people cannot use touch to use the device.

12. References

1. Vityarthi
2. <https://youtu.be/pXiQqaMcFNQ?si=hACSJU-UDFhady2f>
3. https://youtu.be/bIULp8Dxe74?si=EjRB4_XczosMzE08
4. <https://youtu.be/9iEPzbG-xLE?si=JxWwzktx0wWwiJNc>
5. https://youtu.be/vJWzH_2F64g?si=YhUBcElrTnC5iO3g